
CAPTUREE
every atom counts.

recognise this?



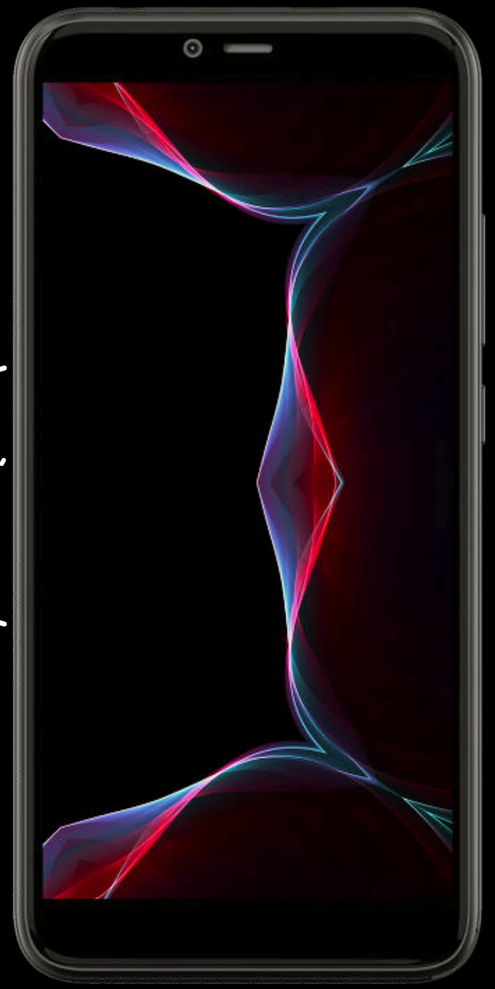
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here is what you don't see.

160mg Neodymium (Nd)

90mg Silver (Ag)

36mg Gold (Au)



70mg Cobalt and Molybdenum

900mg Tungsten (W)

33g Iron (Fe)

7g Chromium (Cr)

13g Silicon (Si)

university of plymouth, 2019

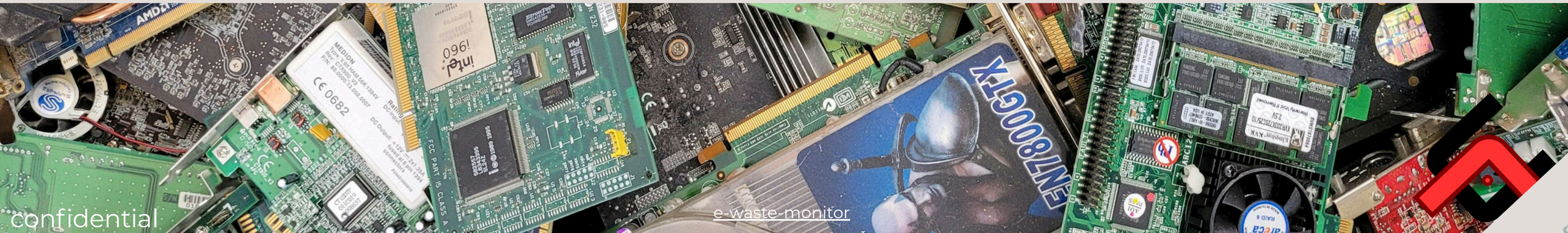
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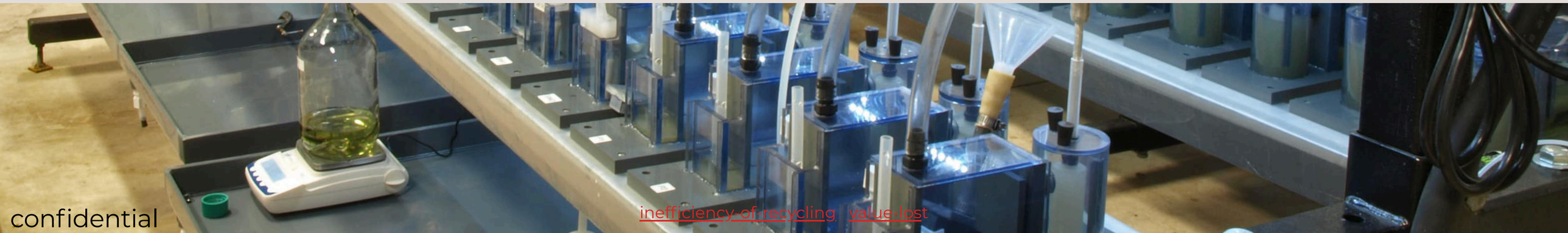
consumption has a cost.

Every year, the world produces over **62 M tonnes of electronic waste** (e-waste), of which **32 M tonnes of critical metals** contained within.



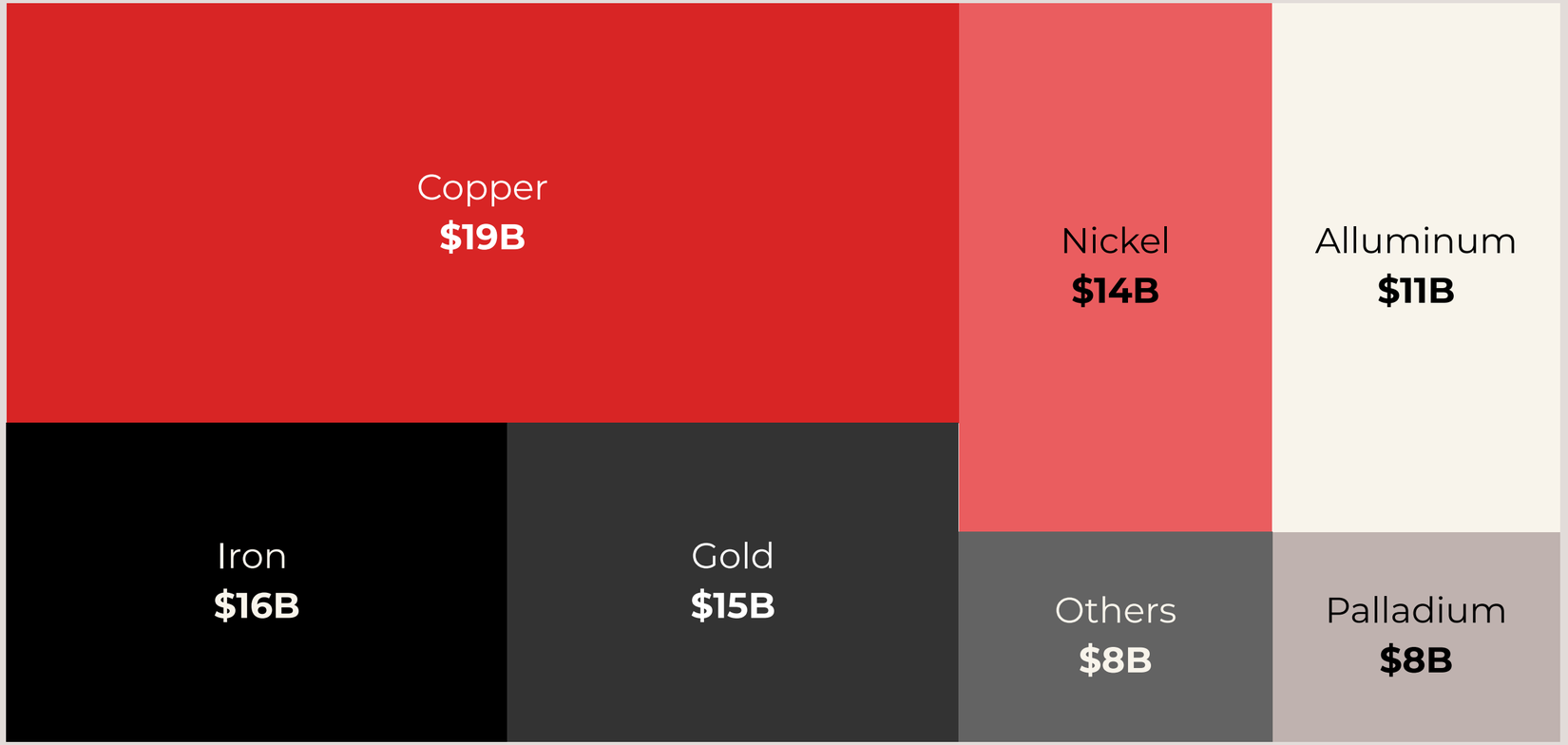
<1% of rare metals recycled.

critical raw materials are dispersed in the bulk waste, and profits do not justify the **extreme costs** and **heavy capex** for the recovery processes implementation.



e-waste is a literal gold mine.

91B
USD



NDA signed

Market capitalization of **€5.38 billion**

Market capitalization of **€7.65 billion**

NDA signed

We talked with 7 European recyclers and OEM industry stakeholders.

Chemical industry groups are focusing R&D on **REE separation**, and they are **ready to pay** for an efficient solution to their problem.

MAGNET FAB ENHANCED MICROSYSTEMS

SOLVAY

SAINT-GOBAIN

Cyclic Materials™

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how we fix this issue

1. better recovery efficiency.

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how we fix this issue

1. better recovery efficiency.

2. higher selectivity.

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how we fix this issue

1. better recovery efficiency.

2. higher selectivity.

3. lower capex.

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CAPTUREE

we offer precision separation technolgies for recovery of critical raw materials.

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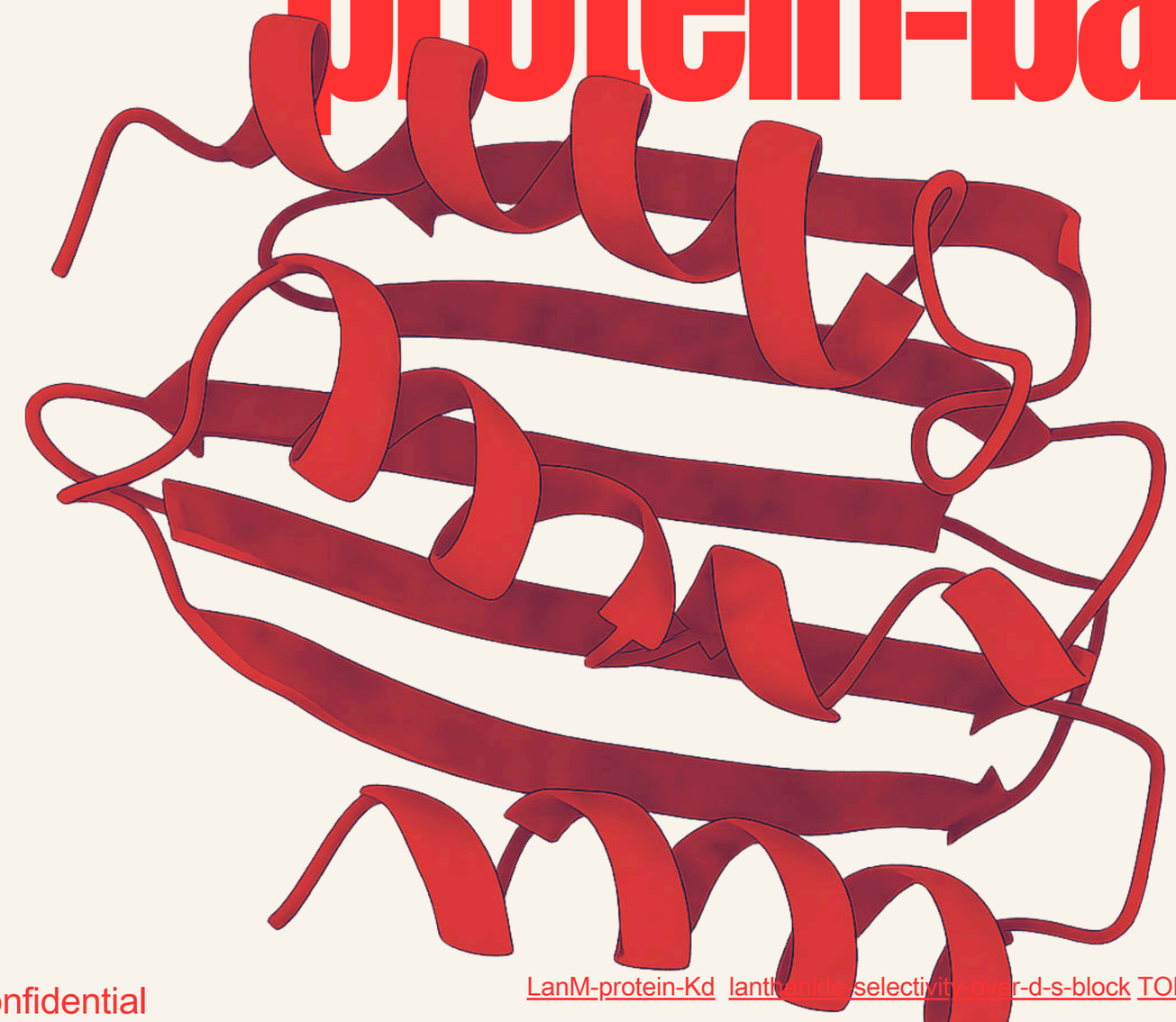
introducing

the CaptuREE catridge.

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plug-&-play

protein-based filter



Engineered proteins

superior metal selectivity.

no organic solvent

vs. TODGA, kerosene-based chemistry

single-pass reocovery + regeneration

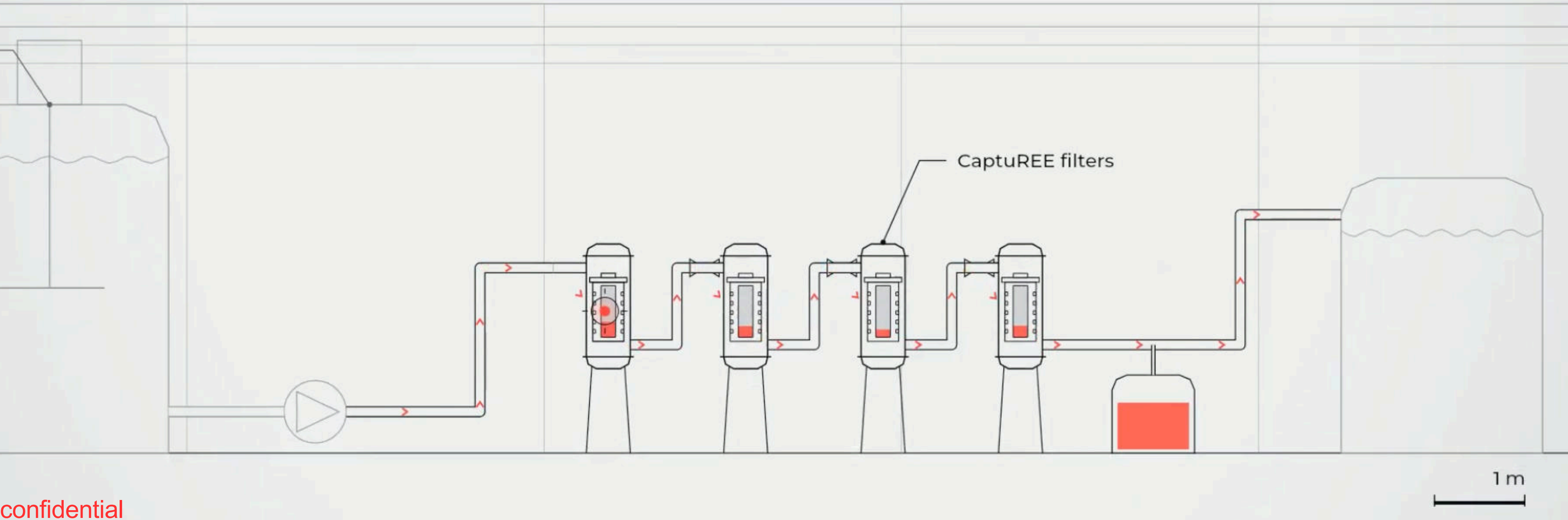
vs. 96% over 5 cycles TODGA → 5× fewer cycles

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LanM-protein-Kd lanthanide-selectivity-over-d-s-block TODGA:DODGA-benchmarks. ree-costs-projections

built to scale.

from a single catridge to **industrial filters** that **plug into existing plants.**



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market size.

CAGR of metal recovery market from e-waste ranges from **6.61%-8.74%**.

Dy & Tb loading from magnet waste and other types of electronic waste.

SOM

534 M

USD



(4% penetration)

unrecovered metals, of which the rare earth fraction is effectively 100% untouched.

SAM

63.1 B

USD

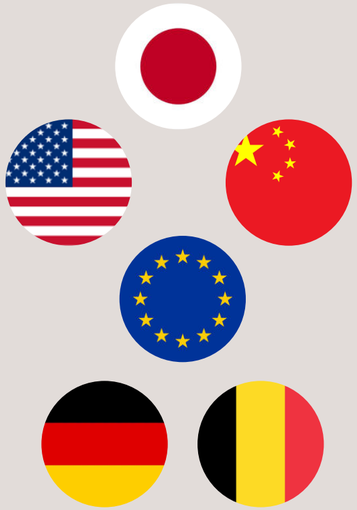


Metal value contained in e-waste generated in 2022.

TAM

91 B

USD



e-waste-market-share. metal-recycling-market-share importance-of-belgium-hydromet global-e-waste-monitor umicore

mixed-revenue business model - metal value share and integration-as-a-service.

We monitor the client's feed **composition** and **recovery potential** through our feasibility analyses, and **share 10-30% of the recovered metal's value** with them, offering a recurring **catridge swap service**.

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	customisable platform	Single REE selectivity	feedstock agnosticity	Industrial implementability	sustainability
CAPTUREE	✓	✓	✓	✓	✓
SOLVAY	✗	✗	✗	✗	✗
Alta	✗	✓	✗	✗	⊖
exære	✓	✗	✗	✓	⊖
Maglut Heavy Industries	✗	✗	✓	✓	⊖
ionic rare earths	✗	✗	✗	✗	✗

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the team.



CEO

Emma Sara Castelli
MSc Molecular Life Sciences
UNIL-EPFL



CTO

Julien Vujevic Martinez
MSc Molecular Life Sciences
UNIL-CHUV

mentors & advisors.

Cara Tobin (Partner @ **Clean Energy Ventures**, **Sustainable Development** coach)

Bertrand Klaiber (**Enterpreneur, Tech4Regen director**)

Julian Englert (**CEO @ Adaptyv Bio**)

Rik Segers (**Business Development director @ Route 2050**)

Johannes Sonnenschein (**Founder, mentor @ iGEM startups**)

Irena Milosevic (**Professor in material sciences @ HSEGE**)

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Europe is building new plants to separate rare earths. We give that capability to those who can’t.

We offer **complementarity** to the system, **improving our client’s processes** and **cutting waste managmenet cost**, all-the-while allowing them to recover the 5% they lose.

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our roadmap.

we are currently here

Q3 2026 | Q4 2026 | Q1 2027 | Q2 2027 | Q3 2027 | Q4 2027 | Q1 2028

iGEM startup showcase

System proven and validated

Prototype unveiling

Angel round

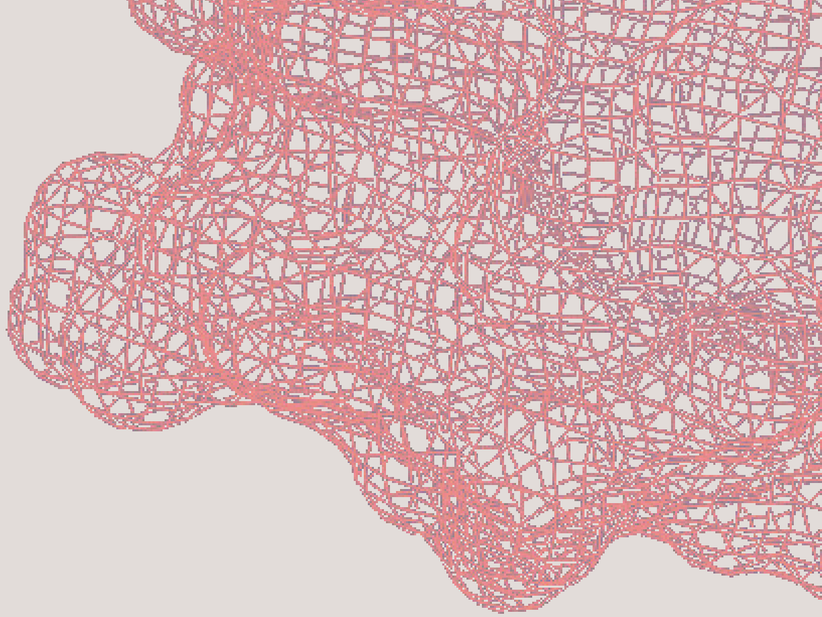
Paid feasibility studies

First industrial pilot

Pre-seed

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building clean raw materials.

powered by **synthethic biology.**